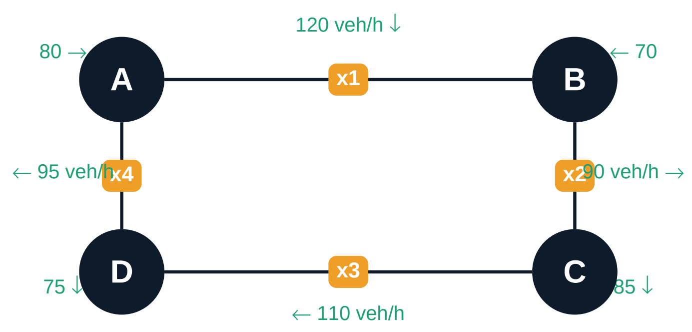
The Problem: Traffic Flow at City Intersections

Real-world application of linear systems

Traffic engineers measure vehicle counts on some road segments but cannot place sensors everywhere. Linear equations let them calculate every unknown flow from the ones they can measure.

NETWORK DIAGRAM



KEY PRINCIPLE

Flow Conservation Law

Vehicles entering any intersection = Vehicles leaving it

One equation per intersection → System of 4 linear equations in 4 unknowns

THE 4 EQUATIONS

Intersection A:	**x1 - x4 = 40**	(120 in + x4 in = 80 out + x1 out)
Intersection B:	**x1 - x2 = -20**	(x1 in + 90 in = 70 out + x2 out)
Intersection C:	**x2 - x3 = -25**	(x2 in + 110 in = x3 out + 85 out)
Intersection D:	**x3 - x4 = -20**	(x3 in + 95 in = x4 out + 75 out)

Solving with Gaussian Elimination

Row reduction on the augmented matrix [A | b]

WHY GAUSSIAN ELIMINATION?

- **Systematic:** works for any size system, no guessing
- **Reveals structure:** a zero row instantly tells us the system is dependent
- **Handles infinite solutions:** identifies free variables automatically

STEP 1 — AUGMENTED MATRIX [A | b]

	x1	x2	x3	x4	|	RHS
Eq A	1	0	0	-1	|	40
Eq B	1	-1	0	0	|	-20
Eq C	0	1	-1	0	|	-25
Eq D	0	0	1	-1	|	-20

STEP 2 — AFTER FORWARD ELIMINATION (upper-triangular)

	x1	x2	x3	x4	|	RHS
Row 1	1	0	0	-1	|	40
Row 2	0	-1	0	1	|	-60
Row 3	0	0	-1	1		
Row 4	0	0	0	0		

Zero row detected!
System is dependent → infinite solutions

STEP 3 — BACK-SUBSTITUTION (set x4 = t)

- Set x4 = t (free parameter, any t > 0)
- Row 3: -x3 + t = -85 x3 = t + 35
- Row 2: -x2 + t = -60 x2 = t + 60
- Row 1: x1 - t = 40 x1 = t + 40

FINAL PARAMETRIC SOLUTION

All four unknowns in terms of t (t > 0, integer, veh/h)

```
x1  =  t + 40
x2  =  t + 60
x3  =  t + 35
x4  =  t
```

— flow on segment A → B
— flow on segment B → C
— flow on segment C → D
— flow on segment D → A

Constraint: all values must be positive → t > 0

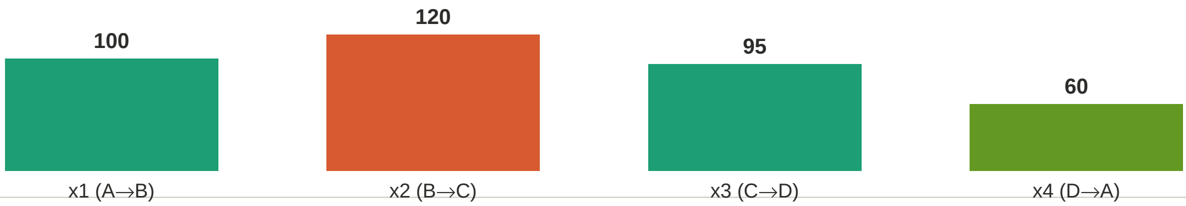
Output, Interpretation & Real-World Use

What does the solution tell a city planner?

EXAMPLE SOLUTION (t = 60 veh/h)

Variable	Segment	Expression	Value (veh/h)	Load
x1	A → B	`t + 40`	**100**	Moderate
x2	B → C	`t + 60`	**120**	HIGHEST
x3	C → D	`t + 35`	**95**	Moderate
x4	D → A	`t`	**60**	Lowest

TRAFFIC LOAD COMPARISON



REAL-WORLD APPLICATIONS

Road planning: Identify which segments need widening or traffic lights

Congestion alerts: Set t bounds from road capacity; flag segments that hit limits

Signal timing: Use flow ratios to set green-light durations at each junction

Emergency routing: Re-solve the system with one road closed to find detour flows

EXTENSION: CONGESTION CHECK (Task 4)

If north inflow at A rises to 180 veh/h:

New system gives x1 = t + 100. Road A→B capacity = 200 veh/h.

Constraint: t + 100 <= 200 => t <= 100.

Congestion occurs on A->B whenever t > 100 veh/h [CONGESTION RISK]

The same approach extends to any city network — more intersections simply add more rows to the matrix.